

4th FTEC WORKSHOP

SPANISH TRAINEESHIP PROGRAMME

Programing of cRIO/FPGA system for High Temperature Superconductors (HTS) Magnets application

Outline

- New SW version of DAQ system.
- SW Architecture
- Cold powering tests Feather-M2 and Results.
- FTEC Programme

New SW version of DAQ system

Main Tasks

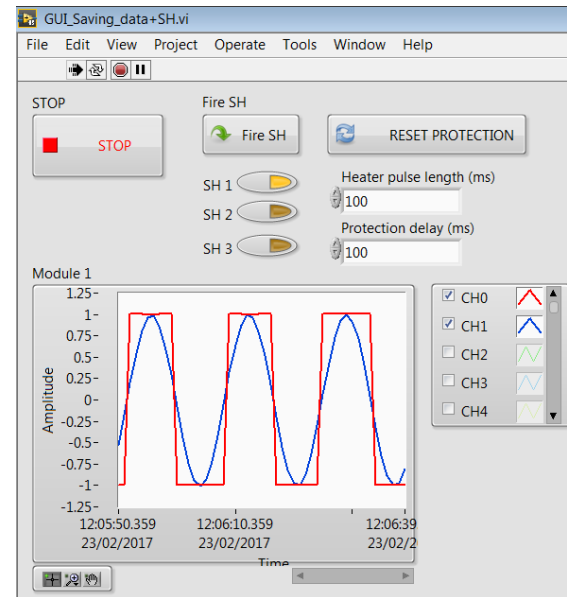
- Continuous acquisition @ 1kHz
- Spot Heater Firing System (SHS)
- Quench Detection System (QDS) @ 10kHz
- RT Monitoring by GUI

Features

- Readout from 34 AI Channels.
- Trigger of Protection Signal.
- Continuous Data Saving
- Data resampling and Conversion.

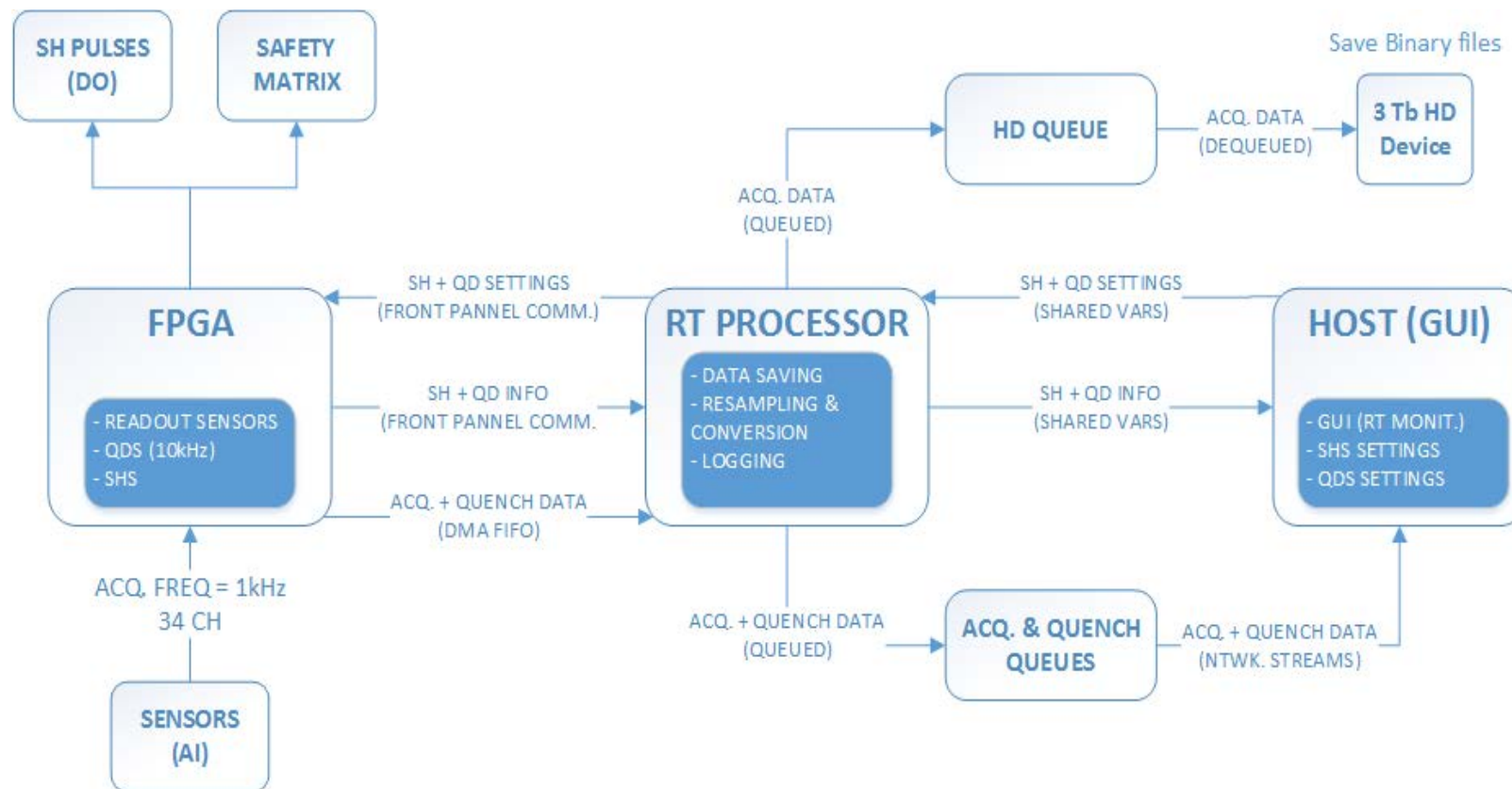


cRIO DAQ system

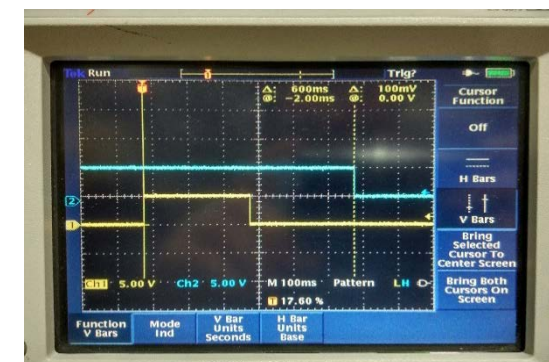


GUI cRIO

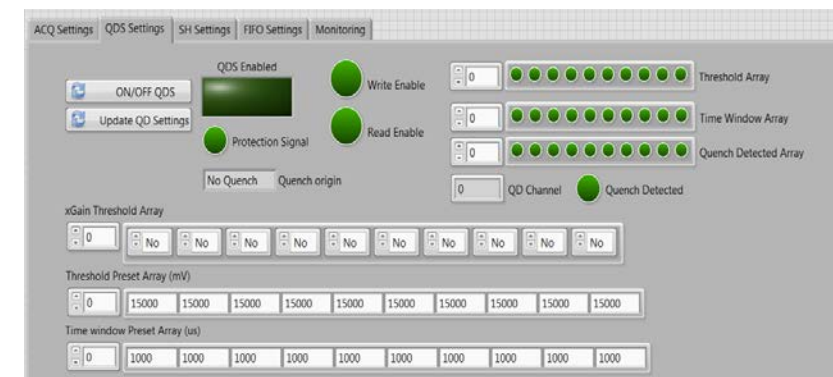
New SW Architecture



cRIO SW Flowchart



SH Pulse + Protection pulse



Quench Detection System

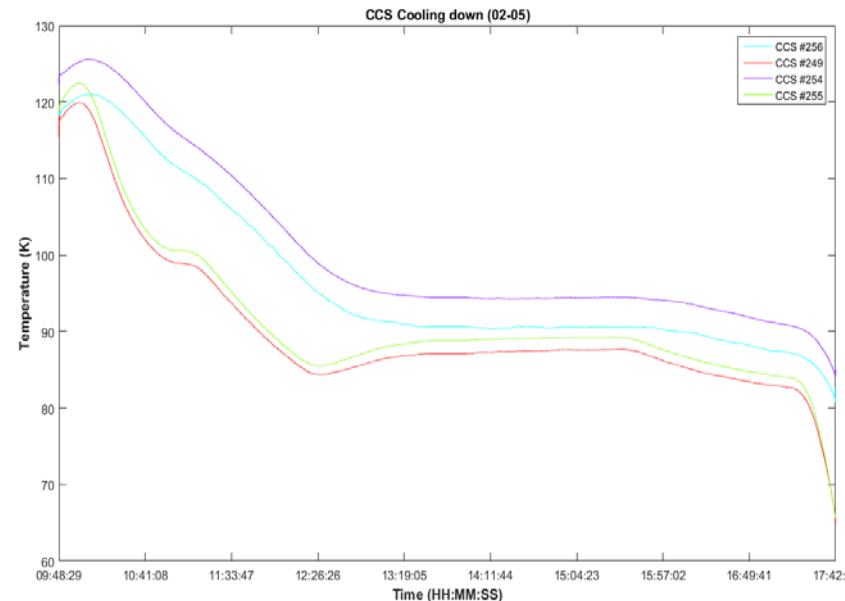
Cold powering tests Feather-M2 and Results.

Feather-M2

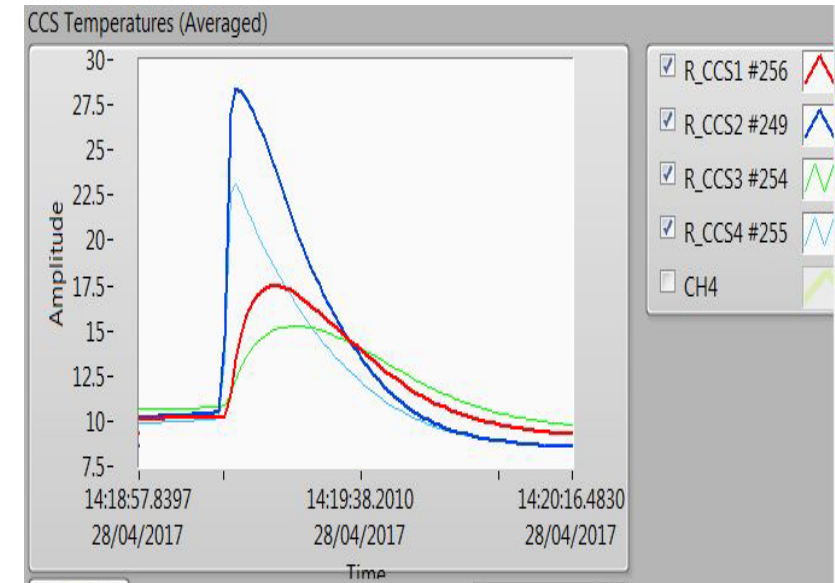
- 2 Coil magnet
- 5x4 Pick-up Coils
- 10 Vtaps
- 4 CCS (Temperature sensors)

Contribution & Results

- RT Monitoring
- CCS Conversion Algorithm
- Data resampling
- Modification of Matlab scripts and plots generation
- **QDS and SHS**



CCS Cooling down



CCS monitoring during a quench

FTEC Programme

- Rewarding experience
- Opportunity to work with HTS Magnets.
- Improvement of Technical skills (LabVIEW & Matlab).
- Learning of specific technical fields (LV FPGAs & NI DAQ)
- Working in an international environment
- Improvement of Language skills
- Aspects to improve
 - Technical reports
 - Help about tax declaration.

Thank you

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